

Due on Tuesday, March 21st, after the Study Break.

If pictures are needed/relevant, please provide them with your solutions.

This problem set is long (3 pages), as appropriate for the Study Break.

The questions marked [**SELF**] are for yourself and need not be submitted.

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1. Two metallic plates are placed parallel and close to each other. They each have area A . They carry opposite charges $+Q$ and $-Q$. Such an arrangement is known as a *capacitor*.

- (a) [**3 pts.**] The plates are large enough that, if we are considering the region between the two plates (far from the plate edges), we can regard each charged plate as an infinite sheet of charge. Show that the magnitude of the electric field in the region between the plates is $E = Q/(A\epsilon_0)$.
- (b) [**2 pts.**] The *capacitance* of a capacitor is the charge on a plate divided by the potential difference between the two plates ($C = Q/V$). If the distance between the plates is d , find the capacitance of the arrangement.

2. Maxwell's 2nd equation asserts that the magnetic field has zero divergence.

- (a) [**3 pts.**] The vector field

$$\mathbf{V}_1 = C_0 x \hat{i} + \frac{C_0}{R} y^2 \hat{j} + 3C_0 R \hat{k}$$

is claimed to describe the magnetic field in some region. (Here C_0 and R are positive constants.) Show that this would violate Maxwell's second equation.

- (b) [**3 pts.**] The vector field $\mathbf{V}_2 = (C_0 R^2) \frac{y \hat{i} - x \hat{j}}{x^2 + y^2}$ is claimed to describe a magnetic field in some region. Is this claim compatible with Maxwell's second equation?
- (c) [**SELF**] What are the dimensions of the quantity R ? (This question means: does R have the units of mass, length, time, charge, or some combination of these?)
Explain your answer.

3. Steady current I flows through an infinitely long straight wire placed along the z axis, from $z = -\infty$ to $z = +\infty$ through the origin.
- (a) [1 pt.] Sketch a top view of the x - y plane, and show the point P with coordinates $(1, 4, 0)$ and the point Q with coordinates $(-3, 2, 0)$. For each of these points, draw the direction of the magnetic field produced by the steady current I .
 - (b) [3 pts.] Calculate the magnitudes of the magnetic field created at those two points (P and Q).
 - (c) [3 pts.] What is the magnitude of the magnetic field at the point $(x_0, y_0, 0)$ on the x - y plane?
 - (d) [4 pts.] At the point $(x_0, y_0, 0)$, find the x -, y -, and z - components of the magnetic field. (Sketching a top view of the x - y plane might help.)
 - (e) [2 pts.] Now consider a point (x_0, y_0, z_0) , not necessarily on the x - y plane. What is the magnitude of the magnetic field at this point? What are the x -, y -, and z - components of the magnetic field at this point?
4. An electron (charge $\approx 1.602 \times 10^{-19}$ C; mass $\approx 9.109 \times 10^{-31}$ kg) moving at speed 100 m/s finds itself in a region of constant magnetic field. The magnetic field points perpendicular to the electron velocity and has strength 10^{-4} Tesla.
- (a) [2 pts.] Find the magnitude of the magnetic force experienced by the electron.
 - (b) [2 pts.] The magnetic force causes cyclotron motion, so that the electron moves around a circular trajectory. Find the radius of the circular trajectory.

5. A particle with charge Q and mass m is placed in a region with constant electric field in the y -direction, $\mathbf{E} = E\hat{j}$, and constant magnetic field in the z direction, $\mathbf{B} = B\hat{k}$. The particle position coordinates are x, y, z , and its velocity components are denoted as v_x, v_y, v_z .

- (a) [4 pts.] Calculate the force $\mathbf{F}$. Write out separately the expressions for its components, F_x, F_y, F_z . (These expressions will involve the quantities Q, B, E, v_x , and v_y .)
- (b) [3 pts.] Using your results for F_x, F_y and Newton's 2nd law, $\mathbf{F} = m\frac{d\mathbf{v}}{dt}$, calculate expressions for $\frac{dv_x}{dt}$ and $\frac{dv_y}{dt}$. You should thus obtain coupled first-order differential equations for $v_x(t)$ and $v_y(t)$.
- (c) [3 pts.] If the particle starts at rest, i.e., if the initial conditions are $v_x(0) = v_y(0) = 0$, then the solutions to the above equations are of the form

$$v_x = \frac{E}{B} - \frac{E}{B} \cos(\omega t) \quad v_y = \frac{E}{B} \sin(\omega t) \quad (1)$$

Substituting these into your first differential equation, find an expression for ω . Substituting these into your second differential equation, show that you find the same expression for ω .

- (d) [4 pts.] The equations (1) are themselves differential equations for $x(t)$ and $y(t)$, since $v_x = \frac{dx}{dt}$ and $v_y = \frac{dy}{dt}$. Assume that the particle starts at the origin, so that the initial conditions are $x(0) = y(0) = 0$. Find the solutions for $x(t)$ and $y(t)$.
- (e) [2 pts.] Sketch $x(t)$ as a function of t and $y(t)$ as a function of t . Sketch the trajectory of the particle in the x - y plane.
6. Two infinite straight wires are placed parallel to each other, at a distance of 3 meters from each other. Each carries a current of 0.5 Amperes.
- (a) [SELF] Look up and report the value of μ_0 (permeability of free space) in SI units. Also calculate μ_0/π .
- (b) [3 pts] If the currents in the two wires are in the same direction, calculate the magnetic field at a point halfway between the two wires.
- (c) [3 pts] If the currents in the two wires are in opposite directions, calculate the magnetic field at a point halfway between the two wires.